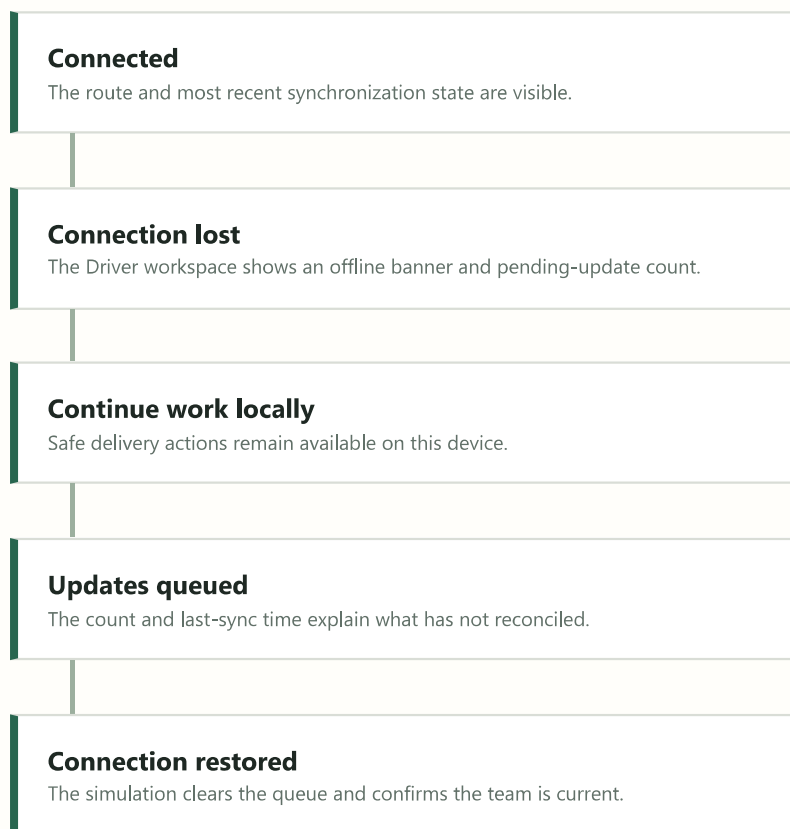


NAMED DEGRADATION SCENARIO

Connection lost during delivery

A driver loses mobile connectivity while executing the route. The work cannot disappear, and the route cannot become ambiguous.

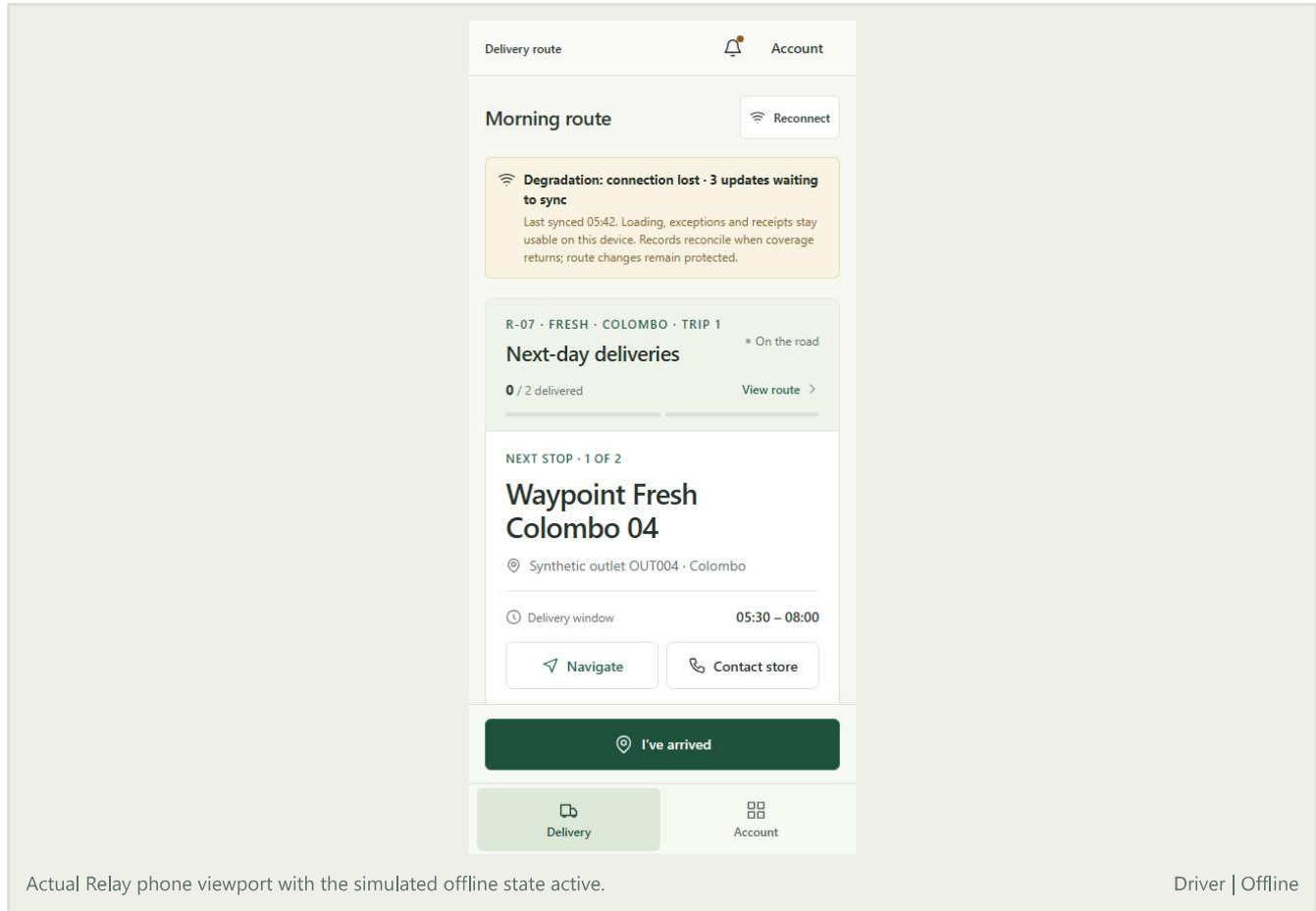
Mobile coverage can drop across hill country, the Kandy corridor, and rural districts. The driver still needs to record arrivals, exceptions, carton checks, and proof of delivery while safely stopped. Relay keeps those actions available in its simulated local state, identifies what is waiting to synchronize, and preserves outstanding stops.

**Design intent**

The degraded state is visible without replacing the driver's current stop. Connectivity changes the status of the record, not the driver's ability to understand the route.

DEGRADED STATE

Offline, with the current stop still usable.



What remains usable

- Current route and next stop
- Arrival recording
- Delivery exceptions
- Receipt details and carton checks
- Loading and issue actions in the Loader view

What the user sees

- Connection lost label
- Pending-update count
- Last synchronization time
- Local-save language
- Outstanding route progress

What is queued

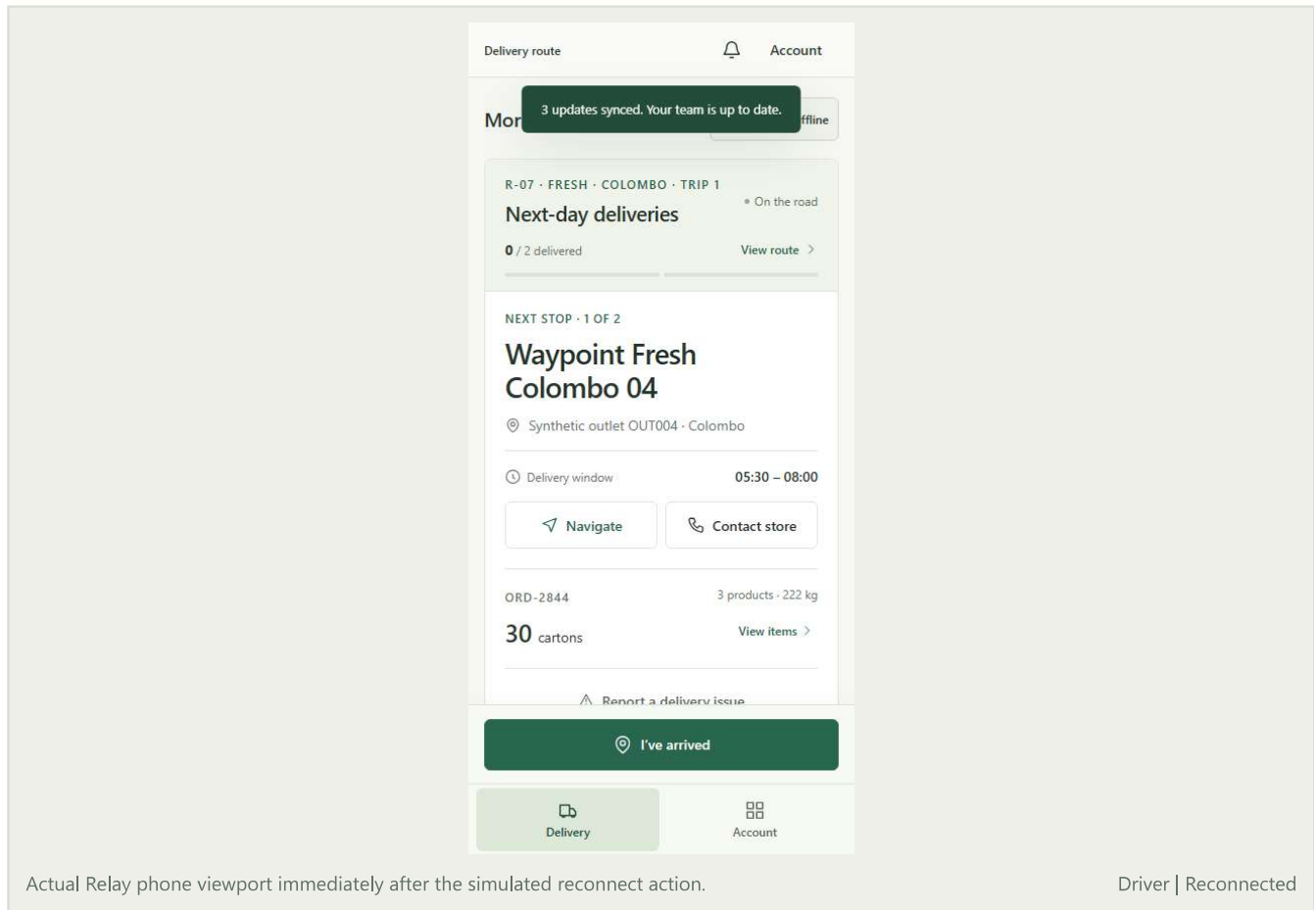
The prototype adds recorded actions to its local pending array while offline. The exact item count updates as the driver works.

What is protected

Dispatch route changes remain unavailable while offline. Relay does not suggest that a remote plan can be safely changed without connectivity.

RECOVERY STATE

Connection restored, queued updates reconciled in the demo.



Recovery feedback

The offline banner disappears, the pending count returns to zero, and a confirmation message states that queued updates were synchronized in the demo workspace.

Continuity

The current route and stop remain in place. Recovery does not force the driver into another screen or hide remaining work.

Prototype simulation, not a production sync system

Relay stores its state in `localStorage` and clears a simulated pending array on reconnect. It does not implement a backend queue, service worker, cross-device reconciliation, conflict handling, retries, or production data durability.

WHY THIS SCENARIO

Connectivity loss crosses every field handoff.

The Challenge Booklet requires work away from the depot to remain usable offline and records to reconcile when connectivity returns.

Operational relevance

The driver is already away from the planning office and warehouse. Losing a connection should not erase the route context, prevent a safely stopped driver from recording the delivery, or make an outstanding stop look complete.

Shared record

Arrival, exception, and POD information later becomes useful to Store and Dispatch. The degraded state makes clear that the record is local and pending before other roles can rely on it.

Safe restraint

The prototype keeps field actions available but protects route changes. It does not imply that every planning action is safe without a connection.

Clear recovery

A count that returns to zero and a specific recovery message give the driver a clear endpoint. Outstanding deliveries still remain visible after reconnection.

Actual limitations

- The offline control is a deliberate simulation.
- State persists only in this browser and on this device.
- No network request is queued or sent.
- No server confirms receipt of a synchronized update.
- No conflict is possible because the prototype has no cross-device state.

Grounded in the implemented Driver and Loader workspaces and the official Challenge Booklet connectivity requirement.